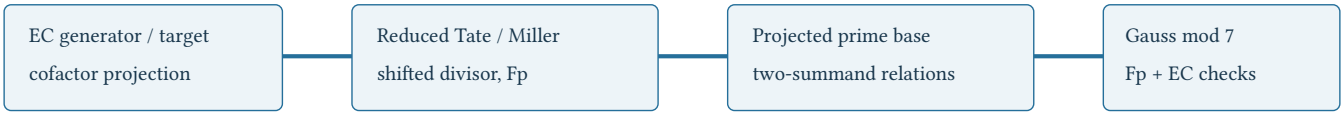
Original C: finite-field reduction diagnostic

Source model icv1-fp61-t2-fefb1f9c. Public synthetic inputs, 17 October 2026. This report is descriptive; no admitted runtime or scaling claim.

The original C model has p=1637958981148213249, a=571607911220311176, b=29329524900995129 and N=1637958981148213248. Native complete factorization gives N=2^26 × 3^20 × 7. Its largest prime subgroup is 7 and embedding degree is 1. The 61-bit field is not a 61-bit prime-order DLP.

Executed diagnostic route: original C, largest prime subgroup r=7



Two factor-base logs and one target log are unknowns. No rho/BSGS fallback. This is not NFS or integer-smoothness IC.

The shifted-divisor Miller implementation computes standard reduced Tate evaluations. Finite tests check all six nonzero control/C scalar inputs and all nontrivial destination generators/targets; these are implementation tests, not a new bilinearity theorem or independent implementation proof. Recovered factor-base logs, the destination answer and the original EC equation are checked. The solver receives neither planted scalars nor factor-base logs.

Original-input arithmetic inventory (not five pairing runs)

Case	Largest prime r	Exact embedding degree	Dense u64 bytes / field element
A	716481758899867	238827252966622	1910618023732976
B	15024241	341460	2731680
C	7	1	8
D	101139959083	101139959082	809119672656
E	87131751841	17426350368	139410802944

Dense coefficient sizes describe that specific layout only. They are not universal lower bounds, impossibility proofs or estimates for other algorithms. The A/B/D/E finite-field routes have not been executed by this diagnostic.

Coverage and retained limitations

Population	Requested targets	Verified	Failed
Original source model	2	2	0
Stored prefix	3738	3738	0

The prefix contains 1,869 stored vertices, 1,024 expanded vertices and 4,903 replayed maps. Its frontier is not exhausted; global stratum populations remain NOT-ESTABLISHED. The separate conservative local certificate is described on page 4. Original C's expected class has 1,934,901,248 vertices. The source-model and prefix populations stay separate, including any repeated target. Two public searches may select the same subgroup point; they are not independent random samples. Other prime-power components are not measured.

Recorded prefix work and elapsed observations

Original C prefix: finite-field work per verified target

n=3738; all stored prefix vertices; two public target searches per model. Not the full C class.

Each row has its own scale. Orange dot: native arithmetic mean. Blue line: observed min–max. No confidence interval.



Relation yield has two explicit meanings: number of collected equations per destination query (several per query are allowed), and fraction of queries with any relation. Identity queries are skipped separately. Relations satisfy the checked destination equation before matrix insertion. Full rank is three.

Metric	n	Min	Mean	Max
query-hit-yield	3738	1.0	1.000000	1.0
relations-per-query	3738	1.0	1.708266	2.5

Native per-target end-to-end elapsed interval: setup/order factorization, point search/projection, pairing, cold factor base, collection/matrix and destination/EC verification. Input file loading, batch serialization and output I/O are outside that per-target interval. Bash process elapsed/user/system logs are separate; peak RSS was not measured because GNU time is absent. All observations are unqualified; there is no external reference or fastest-method claim.

n	Min ns	Mean ns	Max ns
3738	121110	211175.290	3620010

The archive retains both executions, exact binary, source, protocol, fixed prefix DB and input hashes, native summaries and per-target data. The native comparison removes only elapsed time fields, then compares every remaining value, including source/model identities, outcomes, relations and operation ledgers. It is same-binary reproduction, not independent implementation replay. See root-replay-match.json and prefix-replay-match.json for actual checks.

Unpriced work stays explicit: order factorization internals; square-root internals (calls counted); allocation/comparisons; matrix inversions (matrix entry operations counted). Explicit Fp inversions are expanded into their multiplication/square counts: pricing those inversions again would double count. Native EC field counts are separate from the explicit Miller/destination work. There is no common-cost conversion or complete end-to-end GAE claim.

The three complete additional classes are separate datasets. Full A–E graph completion, all-vertex/full-matrix benchmarks, all requested strata, exact representation registration and large-extension finite-field backends remain open. The original graph explorer and fastest-method scoreboards are unchanged: this diagnostic establishes no new complete graph, general algebraic identity or admitted speed comparison.

Per-phase operation distributions (native summary)

Phase	Primitive ledger	n	Min	Mean	Max
EC check	ec-field/invs	3738	0	1.849	3
EC check	ec-field/muls	3738	0	7.716	12
EC check	ec-field/sqrs	3738	0	3.188	5
EC check	ec-group/adds	3738	0	0.509	1
EC check	ec-group/doubles	3738	0	1.339	2
EC check	ec-group/scalar-mults	3738	1	1.000	1
EC check	explicit-fp/additions-subtractions	3738	0	0.000	0
EC check	explicit-fp/exponentiations	3738	0	0.000	0
EC check	explicit-fp/inversions	3738	0	0.000	0
EC check	explicit-fp/multiplications	3738	0	0.000	0
EC check	explicit-fp/squarings	3738	0	0.000	0
EC check	miller-steps	3738	0	0.000	0
EC check	shift-candidates	3738	0	0.000	0
EC check	sqrt-calls-unexpanded	3738	0	0.000	0
FF solve/checks	ec-field/invs	3738	0	0.000	0
FF solve/checks	ec-field/muls	3738	0	0.000	0
FF solve/checks	ec-field/sqrs	3738	0	0.000	0
FF solve/checks	ec-group/adds	3738	0	0.000	0
FF solve/checks	ec-group/doubles	3738	0	0.000	0
FF solve/checks	ec-group/scalar-mults	3738	0	0.000	0
FF solve/checks	explicit-fp/additions-subtractions	3738	0	0.000	0
FF solve/checks	explicit-fp/exponentiations	3738	14	14.000	14
FF solve/checks	explicit-fp/inversions	3738	5	5.000	5
FF solve/checks	explicit-fp/multiplications	3738	322	334.924	347
FF solve/checks	explicit-fp/squarings	3738	533	535.770	538
FF solve/checks	miller-steps	3738	0	0.000	0
FF solve/checks	shift-candidates	3738	0	0.000	0
FF solve/checks	sqrt-calls-unexpanded	3738	0	0.000	0
Pairing	ec-field/invs	3738	14	14.000	14
Pairing	ec-field/muls	3738	52	52.000	52
Pairing	ec-field/sqrs	3738	22	22.000	22
Pairing	ec-group/adds	3738	6	6.000	6
Pairing	ec-group/doubles	3738	8	8.000	8
Pairing	ec-group/scalar-mults	3738	0	0.000	0
Pairing	explicit-fp/additions-subtractions	3738	80	84.195	120
Pairing	explicit-fp/exponentiations	3738	28	28.000	28
Pairing	explicit-fp/inversions	3738	26	26.000	26
Pairing	explicit-fp/multiplications	3738	1270	1274.195	1310
Pairing	explicit-fp/squarings	3738	1692	1694.097	1712
Pairing	miller-steps	3738	8	8.000	8
Pairing	shift-candidates	3738	2	4.097	22
Pairing	sqrt-calls-unexpanded	3738	2	4.097	22
Setup	ec-field/invs	3738	152	174.292	511
Setup	ec-field/muls	3738	658	754.127	2213
Setup	ec-field/sqrs	3738	270	309.473	908
Setup	ec-group/adds	3738	36	41.111	116
Setup	ec-group/doubles	3738	118	136.207	403
Setup	ec-group/scalar-mults	3738	4	4.319	9
Setup	explicit-fp/additions-subtractions	3738	4	9.331	42
Setup	explicit-fp/exponentiations	3738	0	0.000	0
Setup	explicit-fp/inversions	3738	0	0.000	0
Setup	explicit-fp/multiplications	3738	4	9.331	42
Setup	explicit-fp/squarings	3738	2	4.665	21
Setup	miller-steps	3738	0	0.000	0
Setup	shift-candidates	3738	0	0.000	0
Setup	sqrt-calls-unexpanded	3738	2	4.665	21

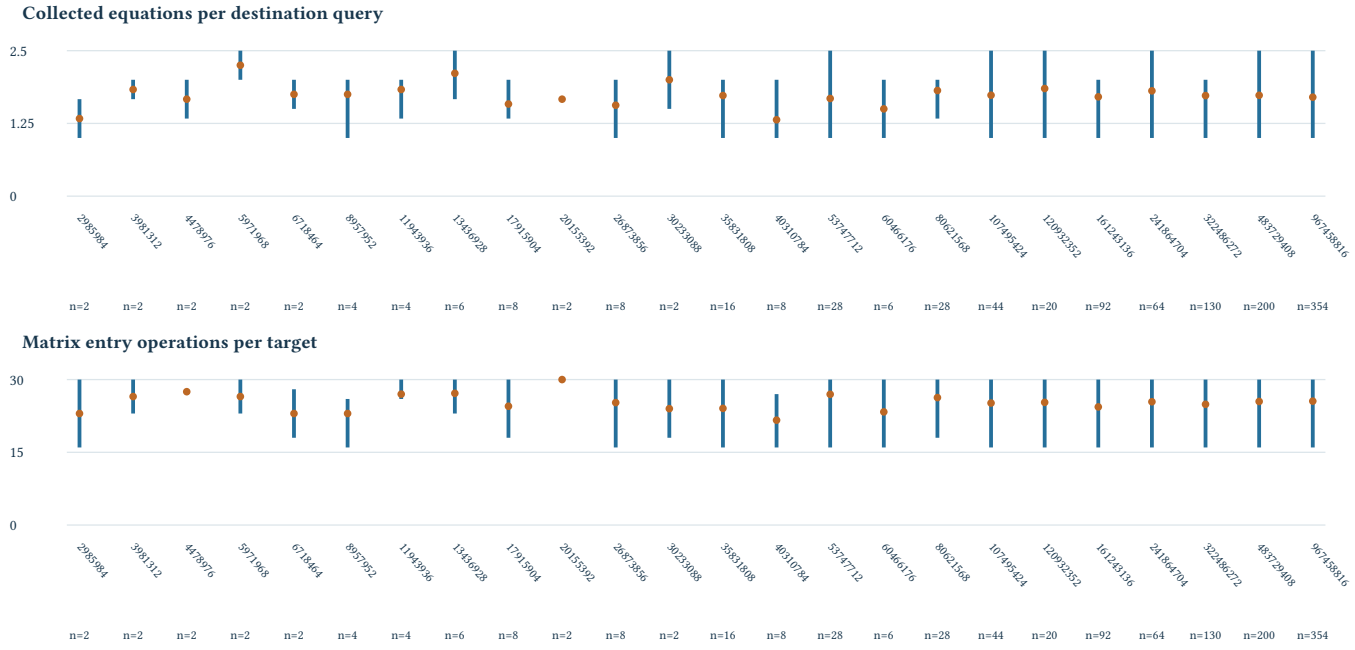
Source: `execution/finite-field-k1-evidence`, `execution/finite-field-k1-summary.json`, `execution/FINITE_FIELD_K1_PROTOCOL.md` and the delivery manifest. Counts cover present verified observations only; failed rows remain in raw data. All raw target and primitive fields are also exported in prefix-observations.csv.

Local conductor bounds: partial coverage of original C

Original C: locally certified conductor strata

24 observed strata; 1034 target records with exact local conductor. Not all 165 C strata.

Orange: native mean. Blue: observed min–max. n is target records; two searches per stored vertex. Ordinal conductor axis.



The local certificate labels 517 vertices across 24 conductor strata, from 2,985,984 to 967,458,816. The source model's exact local conductor is 967,458,816. The other 1352 vertices retain unknown conductor; 2704 target records remain in that separate bucket.

This uses the existing ordinary-volcano floor-distance correspondence. Complete kernel multiplicities identify known floors. Distance to known floors bounds height above; distance to known floors or unexpanded frontier bounds it below. Only equality of finite bounds produces an exact local height/conductor. Unexpanded nodes are possible floors for the bound, never asserted floors. The frozen prefix's global stratum receipt and database are unchanged.

The second labeling implementation passed all checks against the three complete graphs at visibility cutoffs 0, 1, 17, 1024 and all. These are finite calibrations using the same standard volcano theorem and isogeny arithmetic, not a formal proof or independent arithmetic implementation. The full-class population and all 165 strata remain unestablished. Local annotation is study preparation and is not used by the DLP solver or charged to its elapsed interval.

Native per-stratum summaries keep all phases, statuses, min/max/arithmetic means and present-value n. An unknown conductor is a null bucket, not conductor zero or an omitted observation. No causal effect of conductor on cost is inferred from these differing points and sample sizes.